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Task 1: AWS EC2 Instance Creation

Steps Performed

1. Logged in to AWS Management Console.
2. Opened the EC2 Dashboard.
3. Clicked **Launch Instance**.
4. Selected **Ubuntu Server 26.04 LTS** AMI.
5. Chose **t3.small** instance type.
6. Created a new key pair (ec2_key.pem).
7. Created a new Security Group.
8. Allowed the following inbound rules:
 - SSH (Port 22)
 - HTTP (Port 80)
9. Launched the instance.
10. Connected to the instance using SSH.

SSH Command

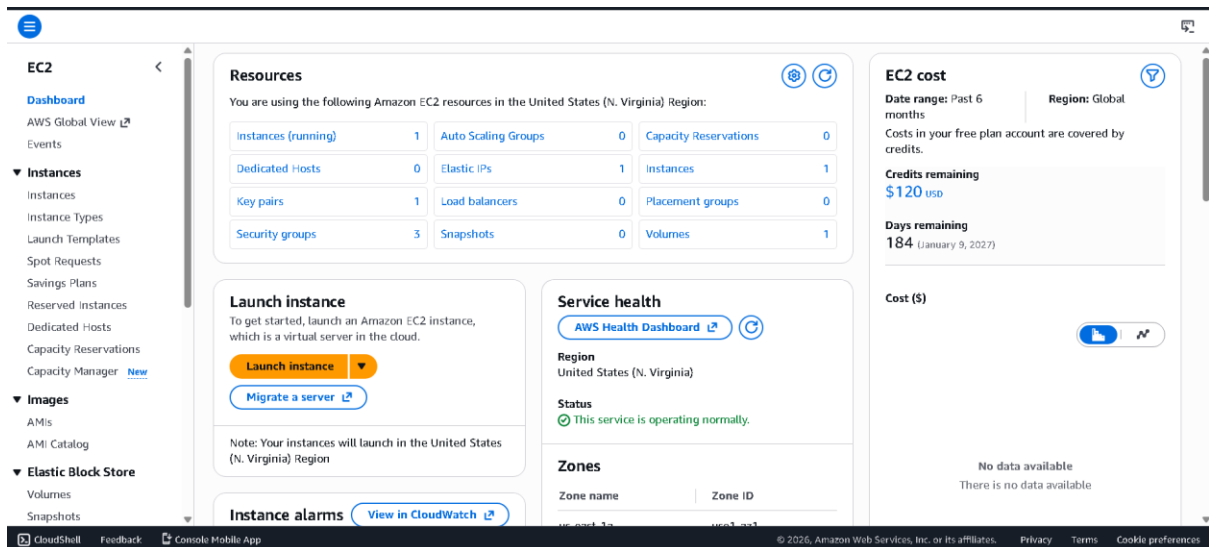
`ssh -i ec2_key.pem ubuntu@3.230.149.94`

AWS Services Used

- Amazon EC2
- Security Groups
- Key Pair

Deliverables

Screenshot 1: EC2 Dashboard



Screenshot 2: SSH Login

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.230.149.94' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Jul  9 14:52:08 UTC 2026

System load:  0.1          Temperature:   -273.1 C
Usage of /:   23.6% of 8.55GB Processes:    120
Memory usage: 13%         Users logged in: 0
Swap usage:   0%          IPv4 address for ens5: 10.0.1.37

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

EC2 Public IP

3.230.149.94

Task 2: Linux Basics

Updating Packages

```
sudo apt update
sudo apt upgrade -y
```

Installing Nginx

```
sudo apt install nginx -y
```

Checking Nginx Status

```
sudo systemctl status nginx
```

Restarting Nginx

```
sudo systemctl restart nginx
```

Checking Disk Usage(df)

```
df -h
```

```
ubuntu@ip-10-0-1-37:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/root        8965896 2347560  6601952  27% /
tmpfs            975468      0    975468   0% /dev/shm
tmpfs            390188      908   389280   1% /run
efivarfs         128          4      120    3% /sys/firmware/efi/efivars
none             1024          0     1024   0% /run/credentials/systemd-journald.service
tmpfs            975472      0    975472   0% /tmp
none             1024          0     1024   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 1012140    97548   845832  11% /boot
/dev/nvme0n1p15 106832     6414   100418   7% /boot/efi
none             1024          0     1024   0% /run/credentials/systemd-networkd.service
none             1024          0     1024   0% /run/credentials/getty@tty1.service
none             1024          0     1024   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            195092      8    195084   1% /run/user/1000
ubuntu@ip-10-0-1-37:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       8.6G  2.3G  6.3G  27% /
tmpfs           953M      0  953M   0% /dev/shm
tmpfs           382M  908K  381M   1% /run
efivarfs        128K   3.1K  120K   3% /sys/firmware/efi/efivars
none            1.0M      0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           953M      0  953M   0% /tmp
none            1.0M      0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13 989M   96M  827M  11% /boot
/dev/nvme0n1p15 105M   6.3M   99M   7% /boot/efi
none            1.0M      0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M      0   1.0M   0% /run/credentials/getty@tty1.service
none            1.0M      0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           191M   8.0K  191M   1% /run/user/1000
```

Checking Memory Usage(free)

```
free -h
```

```
ubuntu@ip-10-0-1-37:~$ free
              total        used        free      shared  buff/cache   available
Mem:           1950940      405980      1079788        2796       638648       1544960
Swap:              0              0              0

ubuntu@ip-10-0-1-37:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       396Mi       1.0Gi       2.7Mi       623Mi       1.5Gi
Swap:              0B              0B              0B
```

Checking Running Processes

ps -ef

```
ubuntu@ip-10-0-1-37:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1      0  0  14:49 ?        00:00:05 /sbin/init
root           2      0  0  14:49 ?        00:00:00 [kthreadd]
root           3      2  0  14:49 ?        00:00:00 [pool_workqueue_release]
root           4      2  0  14:49 ?        00:00:00 [kworker/R-rcu_gp]
root           5      2  0  14:49 ?        00:00:00 [kworker/R-sync_wq]
root           6      2  0  14:49 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7      2  0  14:49 ?        00:00:00 [kworker/R-slub_flushwq]
root           8      2  0  14:49 ?        00:00:00 [kworker/R-netns]
root          10      2  0  14:49 ?        00:00:00 [kworker/0:0H-kblockd]
root          12      2  0  14:49 ?        00:00:00 [kworker/u8:0-events_unbound]
root          13      2  0  14:49 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14      2  0  14:49 ?        00:00:00 [ksoftirqd/0]
root          15      2  0  14:49 ?        00:00:00 [rcu_sched]
root          16      2  0  14:49 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17      2  0  14:49 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18      2  0  14:49 ?        00:00:00 [migration/0]
root          19      2  0  14:49 ?        00:00:00 [kprobe-optimizer]
root          20      2  0  14:49 ?        00:00:00 [idle_inject/0]
root          21      2  0  14:49 ?        00:00:00 [cpuhp/0]
root          22      2  0  14:49 ?        00:00:00 [cpuhp/1]
root          23      2  0  14:49 ?        00:00:00 [idle_inject/1]
root          24      2  0  14:49 ?        00:00:00 [migration/1]
root          25      2  0  14:49 ?        00:00:00 [ksoftirqd/1]
root          27      2  0  14:49 ?        00:00:00 [kworker/1:0H-kblockd]
root          28      2  0  14:49 ?        00:00:00 [kdevtmpfs]
root          29      2  0  14:49 ?        00:00:00 [kworker/R-inet_frag_wq]
root          30      2  0  14:49 ?        00:00:00 [rcu_tasks_kthread]
root          31      2  0  14:49 ?        00:00:00 [rcu_tasks_rude_kthread]
root          32      2  0  14:49 ?        00:00:00 [kauditd]
root          33      2  0  14:49 ?        00:00:00 [khungtaskd]
root          35      2  0  14:49 ?        00:00:00 [oom_reaper]
root          37      2  0  14:49 ?        00:00:00 [kworker/R-writeback]
root          38      2  0  14:49 ?        00:00:00 [kcompactd0]
root          39      2  0  14:49 ?        00:00:00 [ksmd]
root          40      2  0  14:49 ?        00:00:00 [khugepaged]
root          41      2  0  14:49 ?        00:00:00 [kworker/R-kblockd]
root          42      2  0  14:49 ?        00:00:00 [kworker/R-blkcg_punt_bio]
root          43      2  0  14:49 ?        00:00:00 [kworker/R-kintegrityd]
root          44      2  0  14:49 ?        00:00:00 [irq/9-acpi]
root          46      2  0  14:49 ?        00:00:00 [kworker/R-tpm_dev_wq]
root          47      2  0  14:49 ?        00:00:00 [kworker/R-ata_sff]
root          48      2  0  14:49 ?        00:00:00 [kworker/R-md_bitmap]
root          49      2  0  14:49 ?        00:00:00 [kworker/R-md_llbitmap_io]
root          50      2  0  14:49 ?        00:00:00 [kworker/R-md_llbitmap_unplug]
root          51      2  0  14:49 ?        00:00:00 [kworker/R-md]
root          52      2  0  14:49 ?        00:00:00 [kworker/R-edac-poller]
root          53      2  0  14:49 ?        00:00:00 [kworker/R-devfreq_wq]
root          54      2  0  14:49 ?        00:00:00 [watchdogd]
root          55      2  0  14:49 ?        00:00:00 [kworker/R-quota_events_unbound]
root          56      2  0  14:49 ?        00:00:00 [kswapd0]
```

or

top

```
top - 15:01:26 up 11 min, 1 user, load average: 0.00, 0.03, 0.02
Tasks: 118 total, 1 running, 117 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1905.2 total, 1054.0 free, 396.0 used, 624.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1509.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 1990 ubuntu    20   0   11012   5844   3688 R   0.3   0.3   0:00.01 top
   1 root       20   0   25508  16296  11012 S   0.0   0.8   0:02.84 systemd
   2 root       20   0       0       0       0 S   0.0   0.0   0:00.00 kthreadd
   3 root       20   0       0       0       0 S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-rcu_gp
   5 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-sync_wq
   6 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
   7 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-slub_flushwq
   8 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-netns
   9 root       20   0       0       0       0 S   0.0   0.0   0:00.01 kworker/0:0-memcg
  10 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/0:0H-kblockd
  12 root       20   0       0       0       0 S   0.0   0.0   0:00.13 kworker/u8:0-events_power_efficient
  13 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
  14 root       20   0       0       0       0 S   0.0   0.0   0:00.00 ksoftirqd/0
  15 root       20   0       0       0       0 S   0.0   0.0   0:00.07 rcu_sched
  16 root       20   0       0       0       0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
  17 root       20   0       0       0       0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
  18 root       rt    0       0       0       0 S   0.0   0.0   0:00.00 migration/0
  19 root       20   0       0       0       0 S   0.0   0.0   0:00.00 kprobe-optimizer
  20 root      -51   0       0       0       0 S   0.0   0.0   0:00.00 idle_inject/0
  21 root       20   0       0       0       0 S   0.0   0.0   0:00.00 cpuhp/0
  22 root       20   0       0       0       0 S   0.0   0.0   0:00.00 cpuhp/1
  23 root      -51   0       0       0       0 S   0.0   0.0   0:00.00 idle_inject/1
  24 root       rt    0       0       0       0 S   0.0   0.0   0:00.00 migration/1
  25 root       20   0       0       0       0 S   0.0   0.0   0:00.00 ksoftirqd/1
  27 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/1:0H-kblockd
  28 root       20   0       0       0       0 S   0.0   0.0   0:00.00 kdevtmpfs
  29 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
  30 root       20   0       0       0       0 S   0.0   0.0   0:00.00 rcu_tasks_kthread
  31 root       20   0       0       0       0 S   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
  32 root       20   0       0       0       0 S   0.0   0.0   0:00.00 kauditd
  33 root       20   0       0       0       0 S   0.0   0.0   0:00.00 khungtaskd
  34 root       20   0       0       0       0 S   0.0   0.0   0:00.10 kworker/u8:1-events_power_efficient
  35 root       20   0       0       0       0 S   0.0   0.0   0:00.00 oom_reaper
  37 root       0 -20   0       0       0 S   0.0   0.0   0:00.00 kworker/R-writeback
  38 root       20   0       0       0       0 S   0.0   0.0   0:00.02 kcompactd0
  39 root       25   5       0       0       0 S   0.0   0.0   0:00.00 ksmd
```

or

htop

```
0[                                     0.7%] Tasks: 35, 44 thr, 83 kthr; 1 running
1[                                     0.0%] Load average: 0.00 0.03 0.02
Mem[|||||] 229M/1.86G Uptime: 00:11:59
Swp[|||||] 0K/0K

Main 1/0
  PID USER      PRI  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  Command
  1 root       20   0   25508  16296  11012 S   0.0   0.8   0:02.84 /sbin/init
 128 root      -19  -1  50252  14216  12824 S   0.0   0.7   0:00.29 /usr/lib/systemd/systemd-journald
 179 root      -11   0   272M   9364   7292 S   0.0   0.5   0:00.02 /usr/sbin/multipathd -d -s
 185 root       20   0   272M   9364   0 S   0.0   0.5   0:00.00 /usr/sbin/multipathd -d -s
 187 root      -11   0   272M   9364   0 S   0.0   0.5   0:00.00 /usr/sbin/multipathd -d -s
 188 root      -11   0   272M   9364   0 S   0.0   0.5   0:00.00 /usr/sbin/multipathd -d -s
 189 root      -11   0   272M   9364   0 S   0.0   0.5   0:00.00 /usr/sbin/multipathd -d -s
 190 root      -11   0   272M   9364   0 S   0.0   0.5   0:00.03 /usr/sbin/multipathd -d -s
 191 root      -11   0   272M   9364   0 S   0.0   0.5   0:00.00 /usr/sbin/multipathd -d -s
 192 root      -11   0   272M   9364   0 S   0.0   0.5   0:00.00 /usr/sbin/multipathd -d -s
 196 systemd-re 20   0  22460  14244  11508 S   0.0   0.7   0:00.12 /usr/lib/systemd/systemd-resolved
 201 root       20   0  39044  13612   8792 S   0.0   0.7   0:00.42 /usr/lib/systemd/systemd-udev
 595 systemd-ne 20   0  24644  12380  10792 S   0.0   0.6   0:00.08 /usr/lib/systemd/systemd-networkd
 706 root       20   0   2824   2024   1880 S   0.0   0.1   0:00.00 /usr/sbin/acpid
 708 root       20   0   2908   2000   1884 S   0.0   0.1   0:00.03 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
 710 root       20   0   7488   3028   2784 S   0.0   0.2   0:00.00 /usr/sbin/cron -f -P
 711 messagebus 20   0   8748   5316   4328 S   0.0   0.3   0:00.20 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
 721 root       20   0  80796   3888   3592 S   0.0   0.2   0:00.08 /usr/sbin/irqbalance
 724 root       20   0  46888  29756  14604 S   0.0   1.5   0:00.21 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
 726 polkitd    20   0   272M  10060   7508 S   0.0   0.5   0:00.12 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
 731 root       20   0  1985M  39568  26152 S   0.0   2.0   0:00.10 /usr/lib/snapd/snapd
 733 root       20   0  18188   9096   7816 S   0.0   0.5   0:00.11 /usr/lib/systemd/systemd-logind
 735 root       20   0   530M  14984  12620 S   0.0   0.8   0:00.09 /usr/libexec/udisks2/udisksd
 767 root       20   0   530M  14984   0 S   0.0   0.8   0:00.00 /usr/libexec/udisks2/udisksd
 768 root       20   0   8280   2720  2524 S   0.0   0.1   0:00.00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/u
 772 root       20   0  1985M  39568   0 S   0.0   2.0   0:00.03 /usr/lib/snapd/snapd
 773 root       20   0  1985M  39568   0 S   0.0   2.0   0:00.00 /usr/lib/snapd/snapd
 774 root       20   0  1985M  39568   0 S   0.0   2.0   0:00.02 /usr/lib/snapd/snapd
 778 root       20   0  1985M  39568   0 S   0.0   2.0   0:00.06 /usr/lib/snapd/snapd
 779 root       20   0   530M  14984   0 S   0.0   0.8   0:00.00 /usr/libexec/udisks2/udisksd
 780 root       20   0  1985M  39568   0 S   0.0   2.0   0:00.00 /usr/lib/snapd/snapd
 794 root       20   0   8236   2548  2364 S   0.0   0.1   0:00.01 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/u
 796 root       20   0   530M  14984   0 S   0.0   0.8   0:00.00 /usr/libexec/udisks2/udisksd
 812 root       20   0  80796   3888   0 S   0.0   0.2   0:00.00 /usr/sbin/irqbalance
 845 _chrony    20   0  24232  11572  6872 S   0.0   0.6   0:00.06 /usr/sbin/chronyd -n -F 1
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

Deliverables

Screenshot 3: Nginx Installed

```

ubuntu@ip-10-0-1-215:~$ sudo apt install nginx -y
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap nginx-doc ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 60
  Download size: 664 kB
  Space needed: 1876 kB / 4679 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx-common all 1.28.3-2ubuntu1.6 [37.3 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx amd64 1.28.3-2ubuntu1.6 [627 kB]
Fetched 664 kB in 0s (9044 kB/s)
Preconfiguring packages ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

```

Screenshot 4: Nginx Running

```

ubuntu@ip-10-0-1-37:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 14:56:56 UTC; 20s ago
  Invocation: f9df64854d59444a98f1848c1ff3ef55
     Docs: man:nginx(8)
    Process: 1757 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1759 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1788 (nginx)
      Tasks: 3 (limit: 1741)
     Memory: 3M (peak: 7M)
        CPU: 46ms
    CGroup: /system.slice/nginx.service
            └─1788 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─1791 "nginx: worker process"
                └─1792 "nginx: worker process"

Jul 09 14:56:56 ip-10-0-1-37 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server
Jul 09 14:56:56 ip-10-0-1-37 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server

ubuntu@ip-10-0-1-37:~$ sudo systemctl restart nginx
ubuntu@ip-10-0-1-37:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 14:58:01 UTC; 4s ago
  Invocation: 2b2fd9fcf27d42a18a02c72b7c049285
     Docs: man:nginx(8)
    Process: 1884 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1887 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1888 (nginx)
      Tasks: 3 (limit: 1741)
     Memory: 3M (peak: 3.1M)
        CPU: 156ms
    CGroup: /system.slice/nginx.service
            └─1888 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─1889 "nginx: worker process"
                └─1890 "nginx: worker process"

Jul 09 14:58:01 ip-10-0-1-37 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server
Jul 09 14:58:01 ip-10-0-1-37 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server

```

Task 3: Website Hosting

Steps Performed

Created a custom HTML page containing:

- Name
- College
- Branch

- Email
- Current Date

Copied the HTML file into the Nginx web directory.

```
cd /var/www/html
```

```
sudo vi index.html
```

Restarted Nginx.

```
sudo systemctl restart nginx
```

Opened the website using:

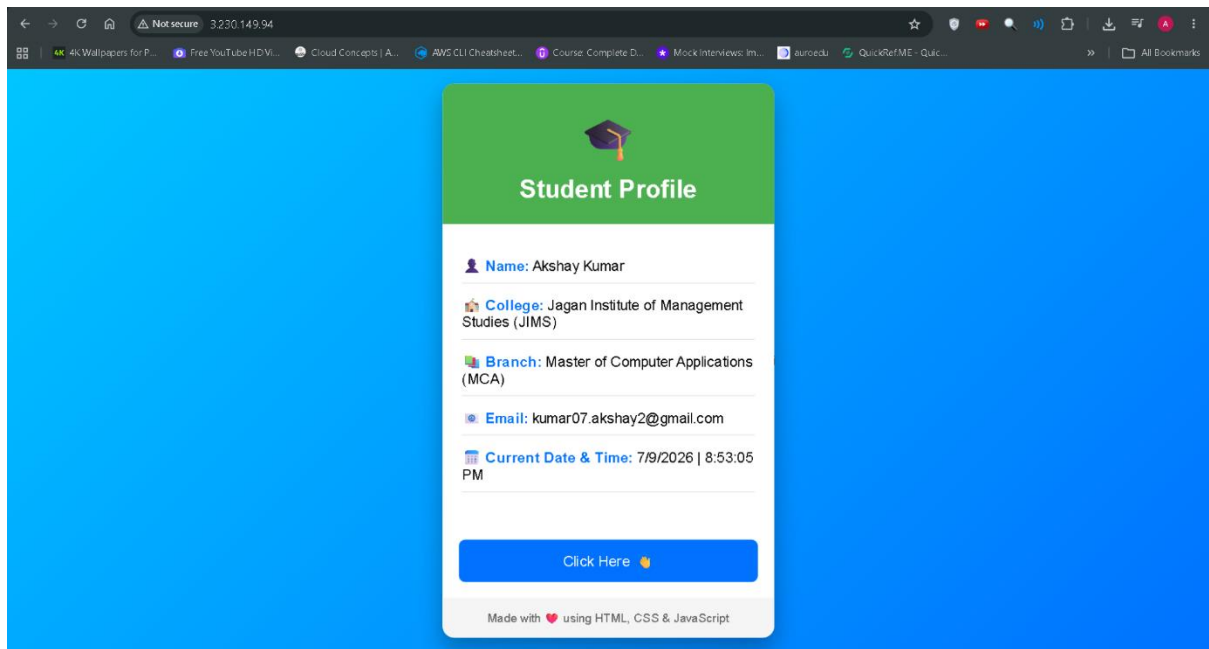
```
http://3.230.149.94
```

Deliverables

Screenshot 5: HTML File

```
<DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title> My Profile</title>
</head>
<body>
<div class="card">
<div class="card-header">
<h2> Welcome</h2>
</div>
<div class="card-body">
<div class="text">
<p>Name</p>
<p>College</p>
<p>Branch</p>
<p>Email</p>
<p>Current Date & Time</p>
</div>
<div class="text">
<p>Name</p>
<p>College</p>
<p>Branch</p>
<p>Email</p>
<p>Current Date & Time</p>
</div>
</div>
</body>
</html>
```

Screenshot 6: Website in Browser



Task 4: Git & GitHub

Steps Performed

Created a GitHub repository.

Initialized Git.

```
git init
```

Added files.

```
git add .
```

Committed the changes.

```
git commit -m "Initial Commit"
```

Connected the GitHub repository.

```
git remote add origin <Repository URL>
```

Pushed the project.

```
git push -u origin main
```

Repository contains:

- index.html

- README.md

README includes:

- EC2 Setup
- Nginx Installation
- Commands Used
- Website Deployment Steps

Deliverable

GitHub Repository

<https://github.com/AkshayKumar-58/AWS-DevOps-Engineer-Intern-Assignment.git>

Task 5: Documentation

AWS Services Used

Amazon EC2

Used to launch a virtual machine for hosting the website.

Security Group

Configured firewall rules to allow SSH (22) and HTTP (80).

Elastic IP (Bonus)

Associated an Elastic IP to ensure a fixed public IP address.

GitHub

Used to store the project source code and documentation.

Linux Commands Used

Command	Purpose
sudo apt update	Update package list
sudo apt upgrade	Upgrade installed packages
sudo apt install nginx	Install Nginx
sudo systemctl start nginx	Start Nginx

Command	Purpose
sudo systemctl restart nginx	Restart Nginx
Sudo systemctl status nginx	Check Nginx status
df -h	Disk usage
free -h	Memory usage
ps -ef	Running processes
top	Monitor system processes
cd	Change directory
ls	List files
vi or nano or vim	Edit files

Problems Faced

Problem 1

Unable to access the website.

Reason

HTTP Port 80 was not allowed in the Security Group.

Solution

Added an inbound rule for Port 80.

Problem 2

Permission denied while editing HTML.

Solution

Used sudo.

```
sudo vi index.html
```

Problem 3

Website showed the default Nginx page.

Solution

Replaced the default index.html with the custom HTML page and restarted Nginx.

Problem 4

Elastic IP association.

Initially the website was accessed using the public IP. Later an Elastic IP was associated so the website remained accessible even after instance restart.

Learnings

Through this assignment, I learned:

- How to launch and configure an AWS EC2 instance.
 - How Security Groups work as virtual firewalls.
 - How to connect to an EC2 instance using SSH.
 - Basic Linux commands and system administration.
 - Installation and management of the Nginx web server.
 - How to host a static website on an EC2 instance.
 - Git and GitHub workflow for version control.
 - Importance of Elastic IP for maintaining a static public address.
 - Basic AWS networking and web hosting concepts.
-

Bonus Task

Elastic IP Association

Successfully allocated and associated an Elastic IP with the EC2 instance.

Benefits:

- Static Public IP
- Persistent website URL
- Easy DNS mapping
- Useful for production environments

Docker Run

Successfully install and run docker run hello-world.

Commands:

```
sudo apt install -y docker.io
```

sudo systemctl start docker

sudo systemctl enable docker

```
ubuntu@ip-10-0-1-37:~$ sudo apt install -y docker.io
docker.io is already the newest version (29.1.3-0ubuntu4.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 60
ubuntu@ip-10-0-1-37:~$ sudo systemctl start docker
ubuntu@ip-10-0-1-37:~$ sudo systemctl enable docker
```

sudo docker run hello-world

```
ubuntu@ip-10-0-1-37:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:96498ffd522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Shell Script to restart Nginx

```
ubuntu@ip-10-0-1-37:~$ cat nginx_restart.sh
#!/bin/bash

sudo systemctl restart nginx
echo "Nginx restarted at $(date)" >> /home/ubuntu/nginx_restart.log
```

Crontab to schedule cronjob for script that runs everyday at 9 A.M.

```
GNU nano 8.7.1 /tmp/crontab.xVGE4Y/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
#
# Restart Nginx every day at 9 A.M.
0 9 * * * /home/ubuntu/nginx_restart.sh
```

Total Time Taken

Task	Time
EC2 Setup	10 Minutes
Linux Commands	10 Minutes
Website Deployment	20 Minutes
GitHub	10 Minutes
Documentation	30 Minutes
Bonus Task	10 Minutes

Total Time: Approximately **2 Hours**

Conclusion

This assignment provided practical experience in deploying a web application on AWS using EC2 and Nginx. It also strengthened my understanding of Linux administration, AWS networking, Git version control, and cloud-based website hosting. The tasks closely resemble real-world DevOps activities, helping me gain hands-on exposure to cloud infrastructure management and deployment workflows.